

Table 1: TruthfulQA: success (%). Bold denotes all tied numerical maxima. Qwen is archival/no-SPR. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	80.6	61.1	94.4	86.1	72.2	91.7	77.8	94.4
Llama 70B	83.3	61.1	91.7	80.6	75.0	83.3	80.6	94.4
Mistral 12B	63.9	61.1	80.6	69.4	63.9	72.7	56.5	86.1
Qwen 7B (archival)	63.2	34.2	68.4	58.3	52.6	65.8	45.5	78.9

Table 2: 2WikiMultiHopQA: success (%). Bold denotes all tied numerical maxima. Qwen is archival/no-SPR. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	86.7	76.7	93.3	78.3	76.7	66.7	80.0	96.7
Llama 70B	96.7	91.7	96.7	95.0	95.0	93.3	90.0	96.7
Mistral 12B	91.7	86.7	95.0	90.0	86.7	89.8	92.2	100.0
Qwen 7B (archival)	81.7	80.0	91.7	85.0	76.7	85.0	90.9	90.0

Jazz results review: scope 16

Across 16 cells, INTRO-SPECTER records 757/866 successes (87.41%). On 866 matched observations, the IS-Reflexion difference is +2.08 pp (87.41% versus 85.33%). The cell record against Reflexion is 10 numerical wins, 2 ties, and 4 losses; 0 wins and 0 losses survive Holm adjustment. Qwen rows are archival no-SPR runs; Qwen is unavailable for rerun. Llama 3.1 8B substitutes for Qwen in new E2/E3 experiments. Baselines are archival; paired task/seed keys do not remove provider drift. Missing observations are not zero scores.

Table 3: LongMemEval: success (%). Bold denotes all tied numerical maxima. Qwen is archival/no-SPR. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	63.3	51.7	76.7	63.3	61.7	50.0	55.0	88.3
Llama 70B	43.3	48.3	70.0	45.0	40.0	33.3	30.0	60.0
Mistral 12B	55.0	50.0	73.3	58.3	58.3	53.3	67.9	81.7
Qwen 7B (archival)	70.0	66.7	76.7	66.7	63.3	75.0	25.0	71.7

Table 4: MuSiQue: success (%). Bold denotes all tied numerical maxima. Qwen is archival/no-SPR. Counts and provenance accompany the generated CSV.

	Direct	SR	Rfx	FR	ReAct	SChk	ToT	INTRO-SPECTER
Llama 8B	76.7	66.7	90.0	76.7	66.7	51.7	71.7	95.0
Llama 70B	83.3	78.3	96.7	81.7	81.7	70.0	55.0	93.3
Mistral 12B	71.7	61.7	85.0	71.7	68.3	75.0	68.6	88.3
Qwen 7B (archival)	80.0	63.3	83.3	80.0	65.0	75.0	77.3	85.0

Table 5: Cell outcomes against Reflexion; wins/ties are numerical, loss significance uses Holm adjustment.

Dataset	Wins	Ties	NS losses	Sig. losses
TruthfulQA	3	1	0	0
2WikiMultiHopQA	2	1	1	0
LongMemEval	2	0	2	0
MuSiQue	3	0	1	0

Table 6: TruthfulQA: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	0.062 [+2.8,+25.0] (36)	< .001* [+19.4,+50.0] (36)	1.000 [+0.0,+0.0] (36)	0.250 [+0.0,+19.4] (36)	0.008 [+8.3,+25.0] (36)
Llama 70B	0.125 [+2.8,+22.2] (36)	< .001* [+19.4,+50.0] (36)	1.000 [+0.0,+8.3] (36)	0.062 [+2.8,+25.0] (36)	0.016 [+8.3,+25.0] (36)
Mistral 12B	0.021 [+5.6,+38.9] (36)	0.022 [+8.3,+41.7] (36)	0.625 [-5.6,+16.7] (36)	0.031 [+5.6,+30.6] (36)	0.008 [+8.3,+25.0] (36)
Qwen 7B (archival)	0.031 [+5.3,+28.9] (38)	< .001* [+26.3,+60.5] (38)	0.219 [+0.0,+23.7] (38)	0.016 [+8.3,+33.3] (36)	0.002* [+13.2,+28.9] (38)

Table 7: 2WikiMultiHopQA: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	0.031 [+3.3,+18.3] (60)	< .001* [+10.0,+30.0] (60)	0.625 [-3.3,+10.0] (60)	< .001* [+8.3,+28.3] (60)	< .001* [+10.0,+30.0] (60)
Llama 70B	1.000 [+0.0,+0.0] (60)	0.250 [+0.0,+11.7] (60)	1.000 [+0.0,+0.0] (60)	1.000 [-3.3,+8.3] (60)	1.000 [+0.0,+0.0] (60)
Mistral 12B	0.062 [+1.7,+16.7] (60)	0.008 [+5.0,+23.3] (60)	0.250 [+0.0,+11.7] (60)	0.031 [+3.3,+18.3] (60)	0.008 [+5.0,+23.3] (60)
Qwen 7B (archival)	0.062 [+1.7,+16.7] (60)	0.031 [+3.3,+18.3] (60)	1.000 [-5.0,+0.0] (60)	0.375 [-1.7,+13.3] (60)	0.008 [+5.0,+23.3] (60)

Table 8: LongMemEval: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	< .001* [+13.3,+36.7] (60)	< .001* [+25.0,+48.3] (60)	0.065 [+1.7,+21.7] (60)	< .001* [+13.3,+36.7] (60)	< .001* [+13.3,+36.7] (60)
Llama 70B	0.006 [+6.7,+28.3] (60)	0.092 [+0.0,+23.3] (60)	0.210 [-23.3,+1.7] (60)	0.004* [+6.7,+25.0] (60)	0.002* [+6.7,+25.0] (60)
Mistral 12B	< .001* [+16.7,+38.3] (60)	< .001* [+20.0,+43.3] (60)	0.227 [-1.7,+18.3] (60)	< .001* [+13.3,+35.0] (60)	< .001* [+13.3,+35.0] (60)
Qwen 7B (archival)	1.000 [+0.0,+5.0] (60)	0.453 [-3.3,+13.3] (60)	0.250 [-11.7,+0.0] (60)	0.250 [+0.0,+11.7] (60)	0.125 [+0.0,+5.0] (60)

Table 9: MuSiQue: exact raw p , paired-bootstrap 95% CI (pp), and paired n in parentheses. *: Holm-adjusted $p < .05$ within the baseline family; sign of the CI determines direction.

	Direct	SR	Rfx	FR	ReAct
Llama 8B	< .001* [+10.0,+28.3] (60)	< .001* [+16.7,+40.0] (60)	0.375 [-1.7,+11.7] (60)	0.003* [+8.3,+30.0] (60)	< .001* [+10.0,+28.3] (60)
Llama 70B	0.070 [+1.7,+20.0] (60)	0.012 [+5.0,+25.0] (60)	0.625 [-10.0,+3.3] (60)	0.039 [+3.3,+21.7] (60)	0.039 [+3.3,+21.7] (60)
Mistral 12B	0.013 [+5.0,+28.3] (60)	< .001* [+13.3,+40.0] (60)	0.754 [-6.7,+13.3] (60)	0.021 [+5.0,+28.3] (60)	0.002* [+5.0,+28.3] (60)
Qwen 7B (archival)	0.250 [+0.0,+11.7] (60)	0.002* [+10.0,+33.3] (60)	1.000 [-3.3,+6.7] (60)	0.508 [-5.0,+15.0] (60)	< .001* [+10.0,+28.3] (60)

Table 10: Recorded mean tokens per observed task; ratios use the recorded IS mean as denominator. Token accounting is inherited from the released harness; archival baselines and SPR reruns have different provider snapshots.

Dataset	Direct	ratio	Rfx	ratio	SChk	ratio	ToT	ratio	IS	ratio
TruthfulQA	1193	0.40	2635	0.87	5887	1.95	34337	11.40	3012	1.00
2WikiMultiHopQA	2010	0.42	2911	0.61	10363	2.16	57592	12.02	4791	1.00
LongMemEval	2180	0.44	5938	1.20	12106	2.44	69436	14.00	4958	1.00
MuSiQue	2128	0.42	4038	0.79	11294	2.22	73129	14.37	5088	1.00